

Project 4:

Teaching Machines to Cool Gas:

Neural Network Emulation of Radiative Cooling in Astrophysical Simulations

Bhumika, Cole, Elliott, Faiyaz, Oscar

Members



Bhumika Joshi
Howard University



Cole Lieberenz
University
of Illinois Chicago



Elliott Lewandowski
University of Alaska
Fairbanks



Oscar Perez
College of DuPage



Faiyaz Chowdhury
Wayne State University



Maria Jimenez
Peer Mentor
Elmhurst University

Members



Joshua Stamborski
Lewis University



Tyler Miller
University of Wyoming



Saleh Abdurrahman
Howard University



Yadiel Torres González
University of
Puerto Rico – Mayaguez



Michael DuBray
South Dakota School of
Mines and Technology



Mohammad Altiwainy
Peer Mentor
Wayne State University

Background

How do large Gas clouds cool?

- This question drives star formation!
- We are tracking Hydrogen and Helium ions.
- Particle interactions between Hydrogen and Helium drive the cooling rate.
- Free electrons are stripped from the atoms further cooling the gas.

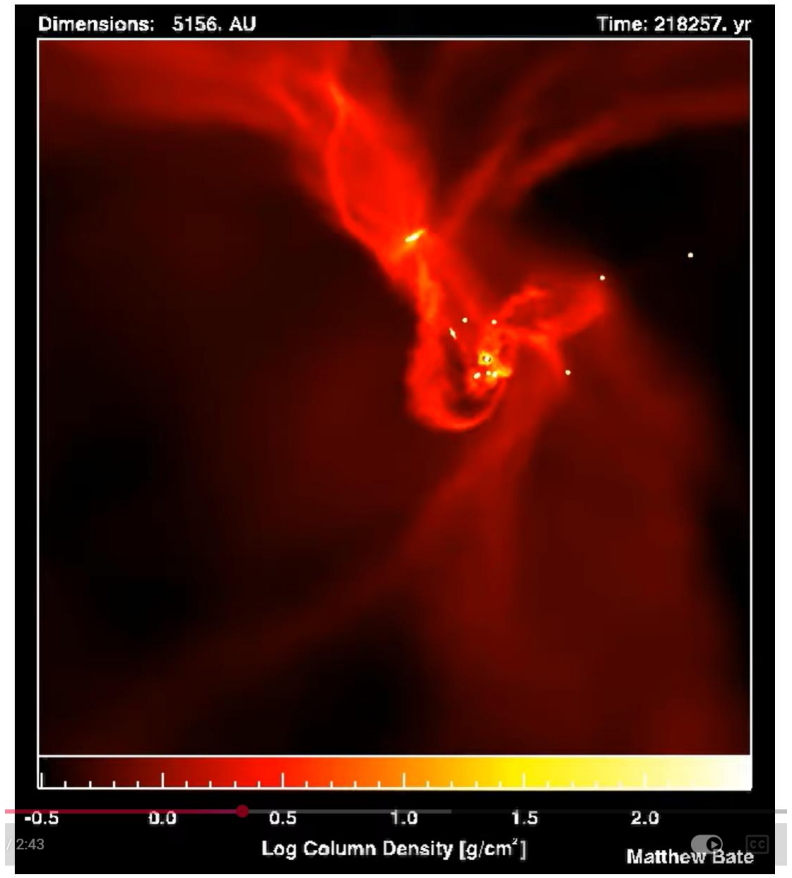
Background

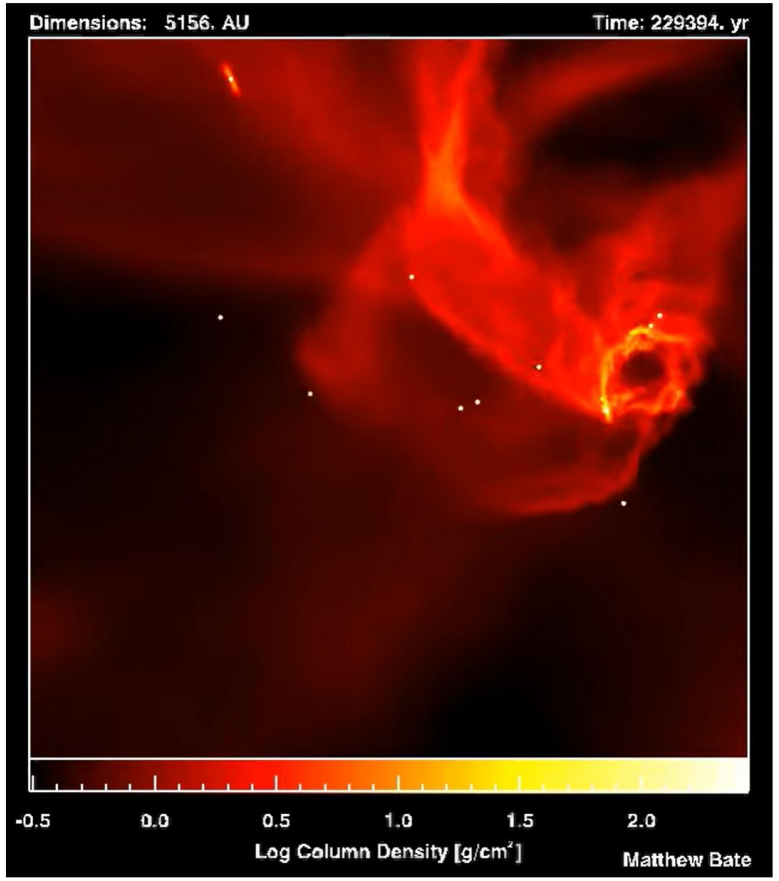
How do large Gas clouds cool?

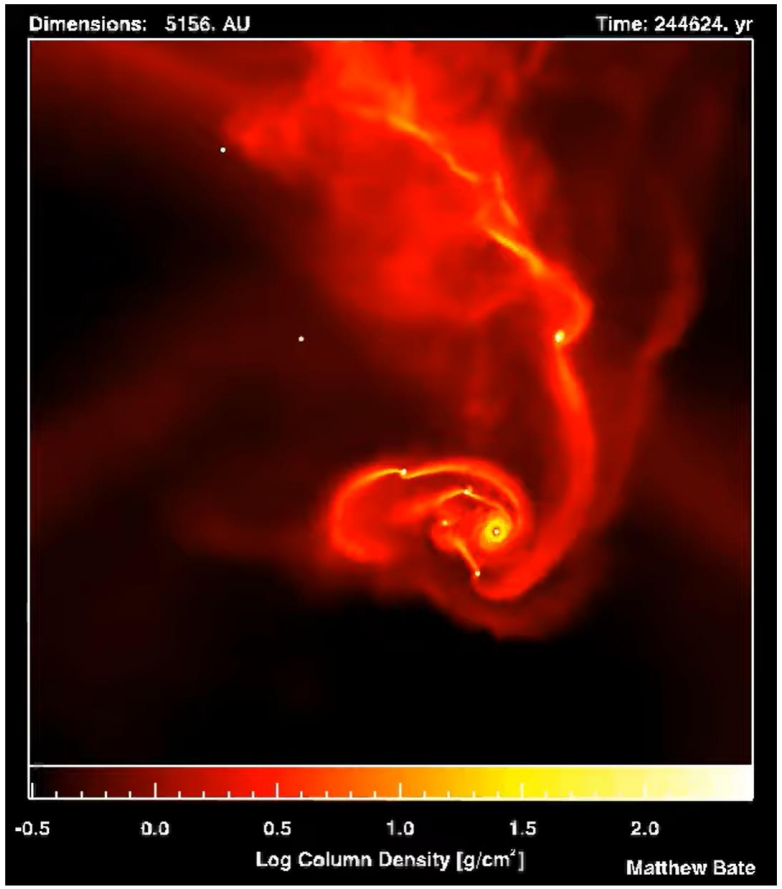
- Each step in each cell of the simulation takes milliseconds.
- One needs a supercomputer when you have 10^{10} cells.
- Faster idea, train a fast imitation neural network!

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Background







Star Cluster Formation via Molecular Cloud Collapse

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Project Description

- Simulate gas cooling
  - Utilize Perlmutter for rapid data generation
- Build a surrogate model
  - Train neural network using simulation data
  - Use to quickly predict gas cooling
- Verify surrogate model results
  - Validate output against simulated results

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Theoretical and Mathematical Methods

Physical Model

- Model **non-equilibrium H/He chemistry** in a gas cell.
- Track hydrogen/helium ionization and **gas temperature**: [xHII ,xHeII ,xHeIII ,T]
- Include **collisional ionization, recombination, and radiative cooling**.

Mathematical Method

- The chemical reactions form a set of **coupled ODEs**: dE/dt =f(y, n_e)
- The system is **stiff** because some reactions occur much faster than others.
- An **implicit Backward Differentiation Formula (BDF) solver** is used to handle these rapidly changing timescales.

From Physics to Neural Networks

- Solve the equations for many **temperature and density** conditions.
- Record the physical results and **time required for each calculation**.
- Use these results to train a **neural-network emulator** that learns the solver's behavior.



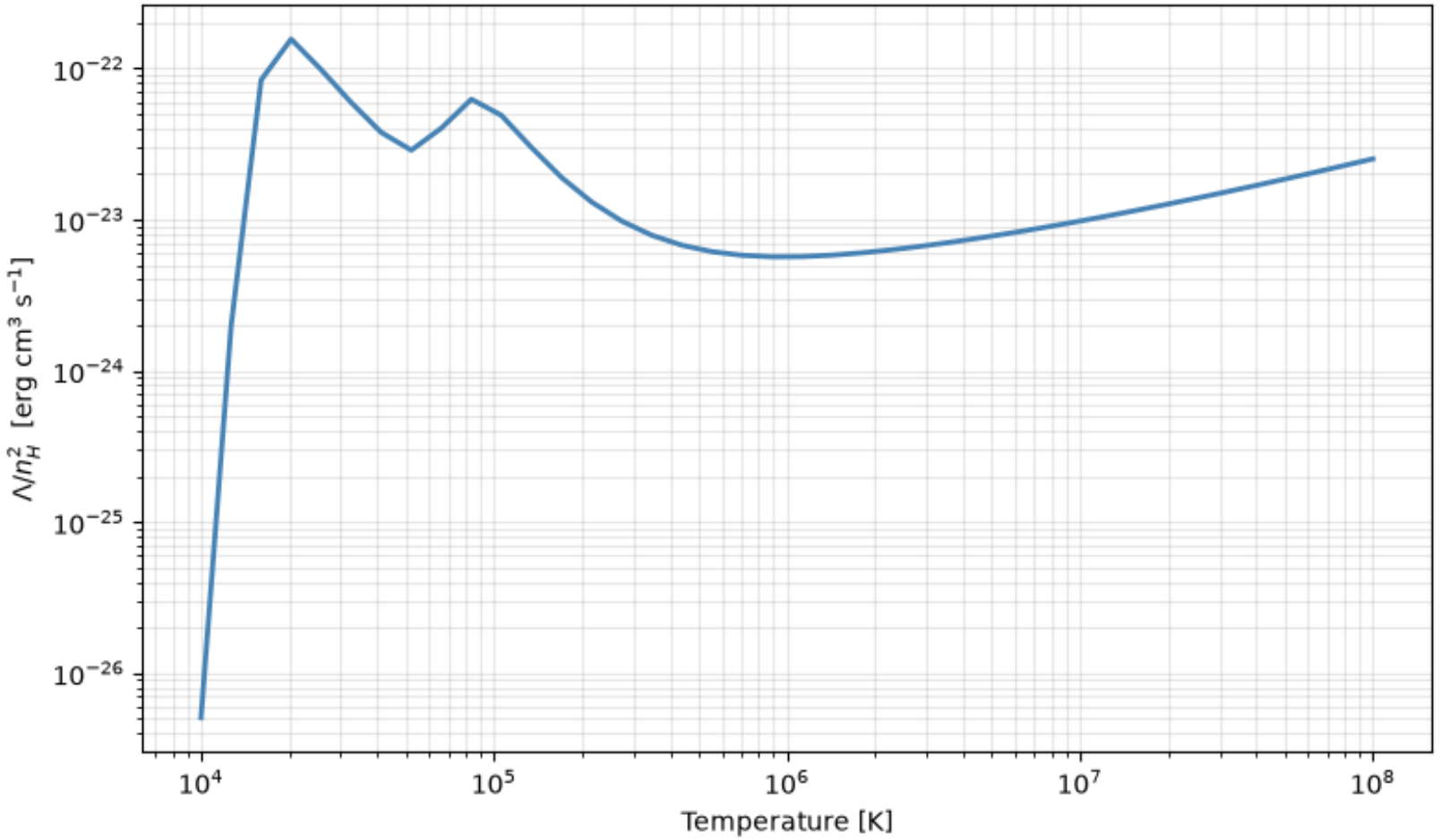
Tools



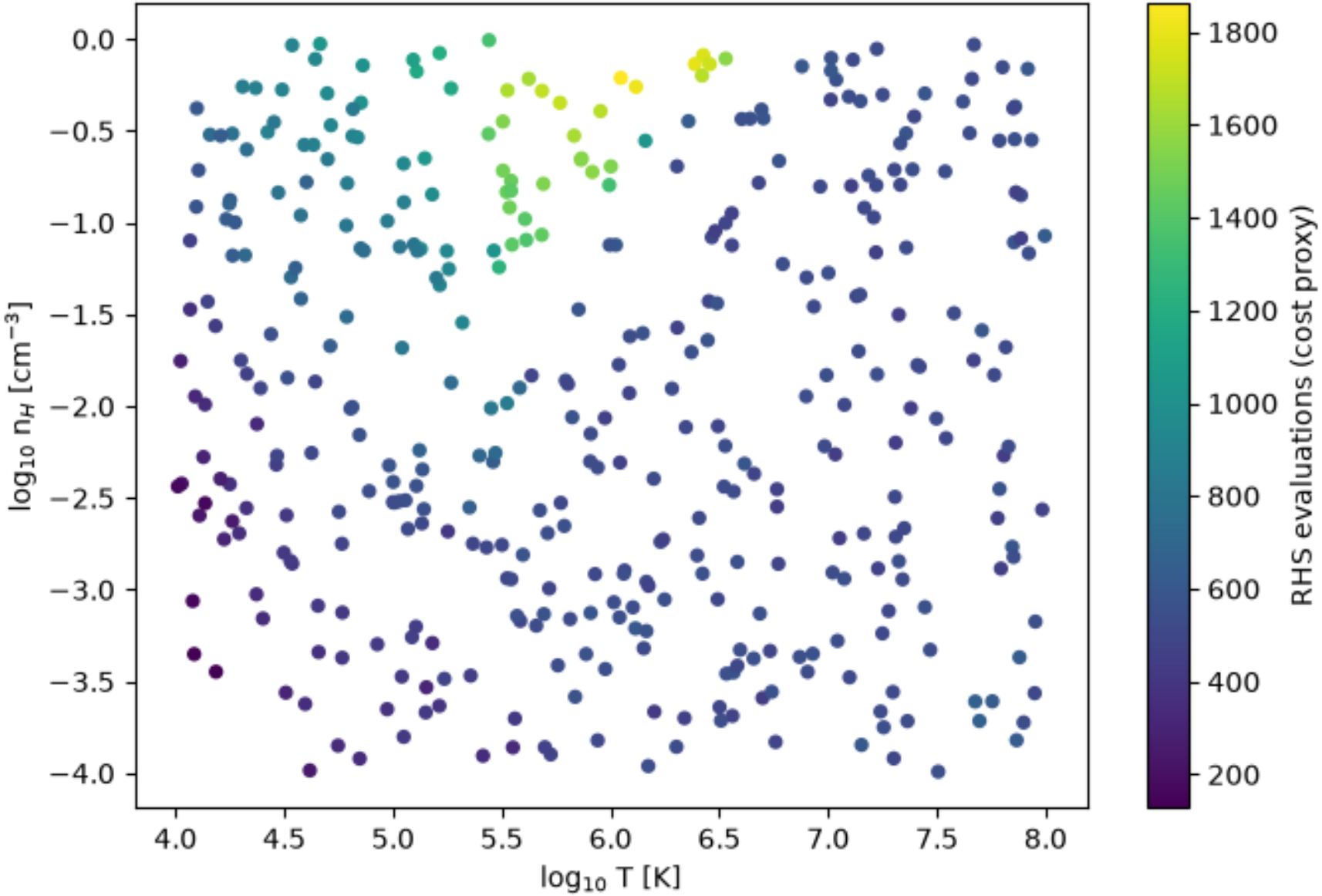
- SLURM
- JupyterNotebook
- Python
- PyTorch
- Numpy
- Scikit-learn

Data and Figures

CIE Cooling Curve, Primordial H/He

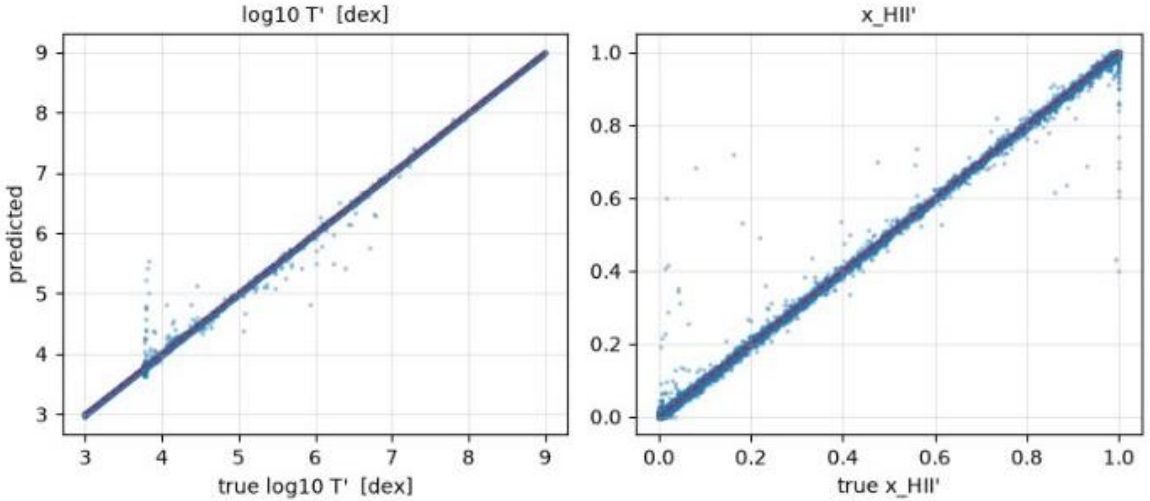


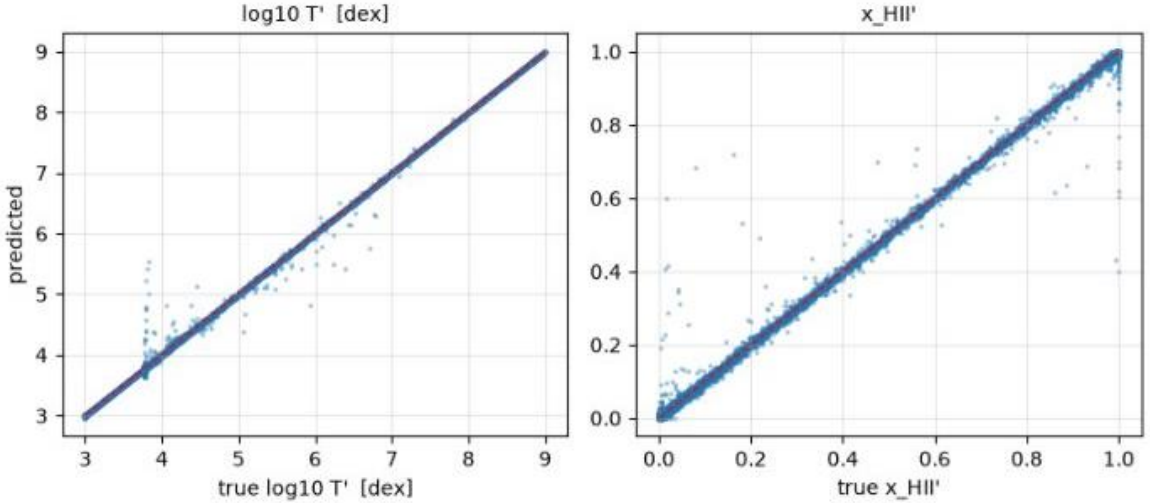
Solver Cost Across Parameter Space



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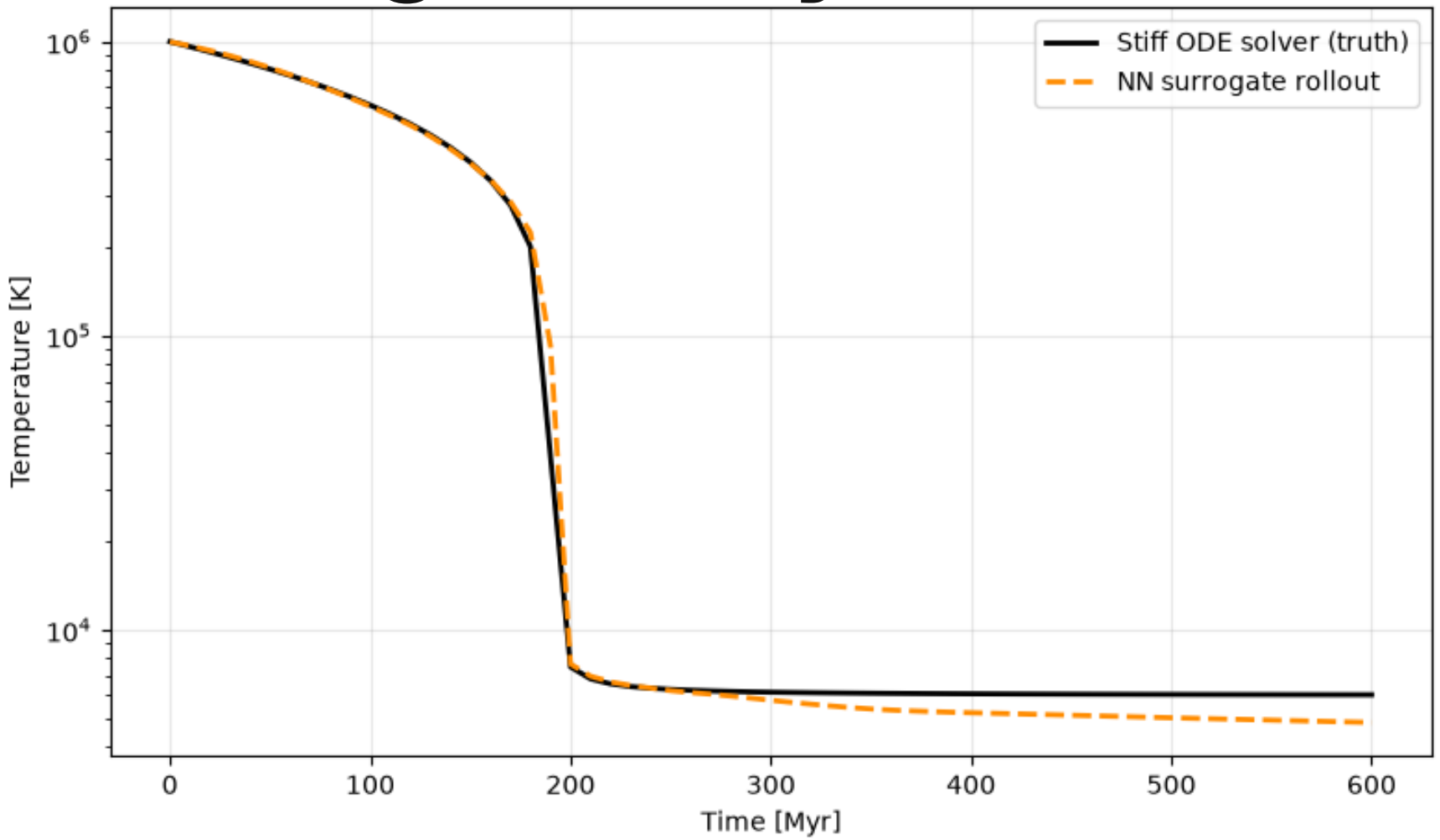
Parity Plots per Target: Model Performance



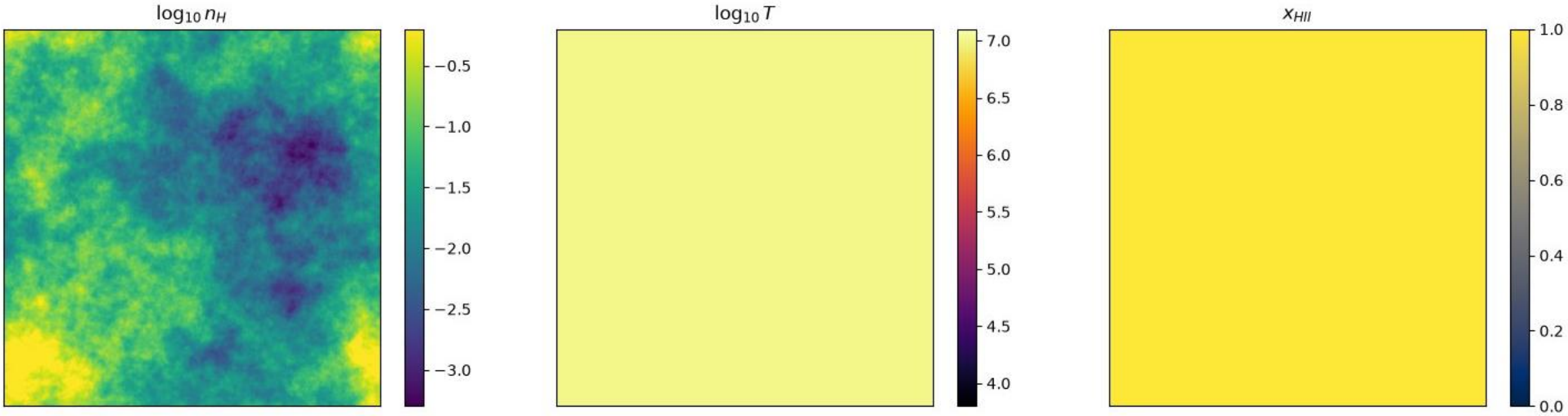


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Autoregressive Rollout: Does The Surrogate Stay on Track?



Application to Simulation



Challenges

General Limitations

- Unfamiliar project terminology
  - Scientific specific
- Technical levels varied among group members
  - Impacted communication/collaboration
- Limited time
  - Tuesday's breakout session cancellation

Challenges

Program Limitations

- Accessing the supercomputer
  - Connection issues
- Data set not large and only slightly non-linear
- Long wait times when training the model
  - Specific chunk in our data was too taking too long

Conclusions and Next Steps

- Residual MLP outperformed plain MLP
- Conceptual overview
- Existing code overview

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Acknowledgments

Project Leads

- **Luis Rangel DaCosta**
  Postdoctoral Researcher, Lawrence Berkeley National Laboratory
- **Snigdaa S. Sethuram**
  Postdoctoral Researcher, Argonne National Laboratory

Peer Mentors

- **Mohammad Altiwainy**
  Wayne State University
- **Maria Jimenez Guillermo**
  Elmhurst University

